

NIS, Serbia

WMO: 133880

Lat: 43.337N

Lon: 21.854E

Elev: 198

StdP: 98.97

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.1	-7.7	-13.2	1.2	-8.7	-10.9	1.5	-6.8	11.4	0.3	9.3	0.1	2.3	320	0.457

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.6	35.2	20.8	33.6	20.4	31.8	20.1	22.4	31.5	21.5	30.5	20.8	29.3	2.3	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.6	14.7	25.5	18.8	14.0	24.7	18.0	13.3	24.0	66.4	31.5	63.5	30.7	60.9	29.5	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.1	7.7	6.5	-13.9	38.2	3.4	2.3	-16.4	39.9	-18.4	41.2	-20.3	42.5	-22.8	44.2		
(5)			-14.3	23.9	3.3	0.9	-16.7	24.6	-18.6	25.1	-20.5	25.6	-22.9	26.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.6	1.1	3.3	7.5	12.6	17.3	21.0	23.1	23.1	18.2	12.9	7.8	2.3		
	DBStd	8.78	4.76	5.20	4.62	4.06	3.56	3.38	3.19	3.17	3.78	4.25	4.59	4.58		
	HDD10.0	951	277	192	104	21	1	0	0	0	0	22	94	241		
	HDD18.3	2582	535	421	336	177	63	13	3	4	48	172	315	496		
	CDD10.0	1889	1	4	27	98	226	331	407	406	247	112	29	3		
	CDD18.3	479	0	0	0	3	30	94	151	151	44	4	1	0		
	CDH23.3	5349	0	0	3	62	342	1017	1681	1699	475	69	2	0		
	CDH26.7	2315	0	0	0	12	83	408	807	827	166	12	0	0		
(14) Wind	WSAvg	2.2	2.2	2.5	2.7	2.4	2.1	2.1	2.1	2.1	2.0	2.0	2.1	2.2		
(15) Precipitation	PrecAvg	615	43	44	44	51	71	74	46	41	49	40	55	57		
	PrecMax	774	87	93	97	105	136	206	93	117	178	84	107	111		
	PrecMin	460	2	7	10	24	21	21	12	6	8	0	18	1		
	PrecStd	84	22	23	27	22	38	39	25	28	43	28	27	31		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.4	19.8	23.5	28.3	31.2	35.0	38.2	38.2	34.2	28.1	22.3	16.9		
		MCWB	9.9	11.5	13.1	16.3	18.5	21.6	20.6	20.5	19.9	17.4	14.7	11.6		
	2%	DB	12.5	16.1	20.8	25.0	28.9	32.9	35.6	35.5	30.9	25.1	19.4	13.9		
		MCWB	8.4	9.7	11.7	14.8	18.2	21.2	20.8	20.7	18.5	16.3	13.2	9.6		
	5%	DB	10.1	13.8	18.2	22.4	27.1	31.0	33.2	33.6	28.1	22.8	17.2	11.2		
		MCWB	7.1	8.6	10.6	13.4	17.7	20.4	20.6	20.1	18.0	15.7	12.2	8.2		
	10%	DB	7.9	11.0	15.9	20.2	25.1	29.0	31.2	31.3	26.0	20.3	15.0	8.9		
		MCWB	5.8	7.4	9.4	12.7	16.9	19.9	20.1	19.8	17.2	14.5	11.0	6.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.2	11.8	13.6	17.2	20.2	23.5	23.1	23.2	20.7	18.5	15.2	11.9		
		MCDB	14.4	18.3	21.8	26.1	27.8	31.1	31.7	32.7	30.6	26.2	20.7	15.6		
	2%	WB	8.7	10.2	12.3	15.6	19.1	22.2	22.2	22.1	19.5	17.2	13.9	10.0		
		MCDB	12.1	15.3	19.2	23.2	26.7	30.6	31.5	31.7	28.3	23.4	18.8	13.1		
	5%	WB	7.2	9.0	11.0	14.4	18.2	21.2	21.5	21.2	18.7	16.0	12.6	8.4		
		MCDB	9.7	13.1	17.2	21.2	25.6	29.2	30.5	30.2	26.5	21.5	16.6	11.1		
	10%	WB	5.7	7.5	9.8	13.2	17.3	20.3	20.7	20.5	17.8	14.9	11.4	6.8		
		MCDB	7.6	10.7	15.4	18.9	24.1	27.9	29.2	29.0	24.6	19.8	14.7	8.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.3	8.9	10.8	11.8	12.1	12.8	13.6	14.1	12.5	11.5	9.5	6.9		
		MCDBR	11.4	14.1	15.6	16.3	15.7	16.0	17.4	18.1	16.9	15.7	13.2	10.9		
	5% WB	MCWBR	7.7	8.5	8.4	8.0	7.1	6.4	6.0	6.2	6.9	7.9	7.6	7.4		
		MCWBR	10.9	12.7	14.2	14.1	14.1	14.6	14.9	15.1	14.6	14.0	12.1	10.2		
(40) Clear-Sky Solar Irradiance	taub		0.336	0.363	0.403	0.461	0.450	0.440	0.442	0.456	0.433	0.392	0.359	0.334		
	taud		2.333	2.250	2.199	2.056	2.148	2.220	2.213	2.174	2.209	2.314	2.362	2.362		
	Ebn at Noon		779	816	829	806	828	836	829	802	788	775	746	744		
	Edh at Noon		83	107	127	159	149	139	139	140	125	100	81	75		
(44) All-Sky Solar Radiation	RadAvg		1.42	2.25	3.38	4.46	5.41	6.16	6.34	5.71	4.04	2.76	1.71	1.22		
	RadStd		0.22	0.31	0.44	0.55	0.53	0.56	0.47	0.51	0.36	0.39	0.20	0.18		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (44 neighbors)	+0.59	N/A	N/A	N/A	+0.46	+0.60	N/A	N/A	N/A	N/A
		+0.61	N/A	N/A	N/A	+0.38	N/A	N/A	-159	N/A	N/A

Nomenclature: See separate page